

HamidReza Saadi Dadmarzi

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Education

Koç University , M.Sc. in Computer Science	2023–Present
Research focus: Password-based authentication systems(MPC), Privacy-preserving ML and Post-Quantum Cryptography.	
Amirkabir University of Technology(Polytechnic of Tehran) , M.Sc. in Applied Mathematics	2017–2020
Research focus: Quantum Error-Correcting Codes and Information-Theoretic Cryptography.	
Amirkabir University of Technology(Polytechnic of Tehran) , B.Sc. in Applied Mathematics	2014–2017
Amirkabir University of Technology(Polytechnic of Tehran) , Minor in Physics	2016–2018

Publications & Research Projects

Publications / Preprints:

- **UpSPA: Secure and Updatable Single Password Authentication** (submitted to ESORICS 2026), ([ePrint version](#), [code](#))
- **RoUpSPA: Robust Updatable Single Password Authentication** (preprint, to be submitted to ACM TOPS)

Selected Research Projects:

- **QuasarSSO: A Post-Quantum Password-Protected SSO System**
- **AEGIS: A Mechanizable Framework for Post-Quantum IoT Authenticated Key Exchange**
- **SpectraDP: Spectral Trade-offs for Communication-Efficient Differentially Private Learning**

Research & Professional Experience

KU-Crypto Lab , Koç University, Research Assistant	2023–Present
Working on cryptographic protocols and secure authentication systems under Prof. Alptekin Küpçü.	
BTL Coin Software Developer	2021–2023
Developed automated crypto-asset management and market analysis tools using Golang and Docker.	
Institute for Research in Fundamental Sciences (IPM)	2020–2021
Worked on quantum machine learning implementations using Qiskit and PyQuil.	

Teaching & Academic Service

- **Teaching assistant for courses in:** Modern Cryptography, Linear Algebra, Computer Networks, Data Structures & Algorithms, Discrete Mathematics, Logic, Probability, PDE, General Physics II, and Calculus I–II at Koç University and Amirkabir University
- **External reviewer for:** IEEE TIFS (2024–present), ACM ASIACCS (2024, 2026), ESORICS (2025, 2026), ACNS (2026), Royal Society Open Science (2024), PeerJ (2026), JISA (2025), and Turkish Journal of Electrical Engineering & Computer Sciences (2025)

Honors & Awards

- Outstanding Teaching Assistant Award, Koç University (2025).
- Exceptional Talent Center recognition for graduate admissions and academic performance.
- Ranked among the top 5% of undergraduate students.

Technical Skills

Programming: Rust, Go, Python, MATLAB
Tools: Docker, L^AT_EX, Qiskit, PyQuil, Lattigo
Areas: Cryptography, SMPC, PQC.

Conferences & Workshops

- Frontiers in Mathematical Science 5th,6th Conference - in IPM(Tehran) - July 2017,2018
- Workshop on Mathematics of Information-Theoretic Cryptography - in [IMS](#), Singapore - September 2016

References

- Prof. Alptekin Küpçü([link](#),[email](#))
- Dr. Erfan Salavati([link](#),[email](#))
- Prof. Masoud Pourmahdian([link](#),[email](#))